

BidKV-MultiNPU: Ownership-Aware Victim Coordination Under Distributed KV Pressure

Advisor-authored storyline draft; implementation and experiments remain student-owned

M0 proposal, August 2026

Abstract

BidKV is an accepted single-device victim-selection policy for KV-pressure LLM serving. Running that policy on several accelerators does not by itself create a multi-device mechanism: a request may have state on several owners, pressure can be asymmetric, and a locally attractive victim may trigger duplicated recomputation, collective delay, or cross-device traffic. This paper draft poses a separate, falsifiable question: when do ownership and topology make independent local victim choices globally costly, and can bounded coordination avoid that cost without global serialization? No MultiNPU implementation, deterministic replay, or hardware result is admitted today. We freeze seven research questions, a six-link NPU claim chain, local/global/oracle baselines, conservation oracles, performance and coordination-overhead gates, negative stops, and a two-week delivery. Until an ownership counterexample and native coordination path exist, the extension remains design-level porting/adaptation, not an NPU mechanism contribution.

1 Introduction

Paged KV memory allows serving engines to reclaim active request state under pressure [3]. BidKV improves the local victim decision by ranking the capacity a victim frees against its estimated disruption cost [1]. The accepted manuscript and its NVIDIA results are the inherited single-device baseline. They contain no evidence that independent choices remain correct when one logical request spans or interacts with several accelerators.

The multi-device extension begins where the local abstraction can fail. A decision made from one device’s pressure may ignore blocks owned elsewhere, collective dependencies, asynchronous ownership updates, recomputation that is charged twice, or transfer induced by moving the decision boundary. Conversely, a global coordinator can cost more than it saves. The contribution therefore cannot be “global BidKV.” It must identify an observable divergence between local and globally accountable choices, implement the minimum metadata exchange that repairs it, and predict when coordination should remain off.

The repository currently provides only that research contract. Historical single-device result directories remain evidence for inherited BidKV and are never relabeled MultiNPU. Dedicated branches identify future carriers but do not admit code or results into this draft.

2 Seven-Question Contract

Q	Frozen answer
1	Problem. Under asymmetric per-device KV pressure and shared request ownership, when does independent local victim selection choose a globally expensive victim?
2	Importance. A bad coordinated decision can free the wrong physical capacity, duplicate recomputation, delay collectives, or serialize the serving path.
3	Hidden assumption and gap. Local BidKV assumes that capacity and disruption are attributable at one decision point. Generic multi-resource fairness [2] does not supply request/KV ownership or recomputation semantics.
4	Mechanism hypothesis. Exchange a bounded ownership-and-cost receipt only at pressure events, reject stale epochs, and coordinate only when predicted global regret exceeds communication uncertainty; otherwise retain local BidKV.
5	Feasibility. Dedicated parent/runtime branches exist, but exact ownership hooks, replay, and coordination code are pending student work.
6	Matched evaluation. Compare independent local BidKV, the strongest non-BidKV local policy, global largest-first, the treatment, and an offline coordination oracle under identical topology, memory, requests, and restarts.
7	Takeaway. A reusable result is a measured boundary identifying when ownership-aware coordination pays, or a negative result showing that its cost dominates on tested topologies.

Table 1: Questions are preregistered obligations, not completed claims.

3 Ownership Model and Hidden Assumption

Let $d \in \mathcal{D}$ index devices. Request i owns physical KV blocks $B_{i,d}$, and a local policy observes pressure P_d . Reclaiming i produces a vector of freed capacity $\mathbf{f}_i = (f_{i,d})$ and a global disruption C_i that includes recomputation, synchronization, and any transfer. A local score $f_{i,d}/\widehat{C}_{i,d}$ is safe only if local ownership and local cost are sufficient statistics for $\mathbf{f}_i$ and C_i . The project tests that assumption rather than declaring it false.

A candidate coordination receipt contains request and generation identity, ownership epoch, per-device blocks, lifecycle state, local pressure, predicted recompute/transfer cost, and monotonic timestamp. A stale or incomplete receipt must not trigger a global action. The proposed coordinator selects at most the minimum victim set needed to satisfy explicit per-device deficits and emits an auditable decision epoch. This is a design hypothesis, not an implementation.

4 NPU Six-Link Claim Discipline

1. **Architecture fact.** The selected TP/DP carrier is expected to distribute request execution and KV ownership, but the precise ownership map and hook identities are still an M0 deliverable.
2. **Invalidated assumption.** The candidate invalid assumption is that each device can price and reclaim victims independently without affecting peers. It has not been invalidated on the pinned carrier.
3. **Real counterexample.** None exists. M0 first requires a deterministic replay case where the local optimum is globally worse; replay would still be simulation, not a hardware counterexample.
4. **Native mechanism.** Ownership receipts and bounded coordination at named vLLM-HUST/Ascend hook points are pending.
5. **Predictive boundary.** Coordination should help only when pressure and ownership are asymmetric, local/global choices diverge, and avoidable global cost exceeds message, synchronization, and staleness cost.
6. **Generalization class.** A verified rule may generalize to serving systems with explicit sharded state ownership; it does not cover replicas with independent KV state or opaque ownership.

The first four links are incomplete. BidKV-MultiNPU is therefore explicitly an **unverified porting/adaptation study** at this head.

5 Mechanism Hypothesis

Coordination is pressure-triggered, not continuously centralized. Each device publishes a bounded receipt for its candidate victims. A decision is accepted only if all required owners share one epoch and the predicted global regret gap exceeds uncertainty plus coordination cost. Otherwise each device uses the inherited local BidKV policy. The coordinator neither changes admission nor compresses KV and never treats a request as released until every owning device acknowledges its physical block release.

This scope excludes routing, placement, migration policy, cache compression, load balancing, and new quality surrogates. It also excludes global fairness as a primary objective: fairness is a safety metric, while the mechanism claim is ownership-correct victim coordination.

6 Evidence Ledger

Asset	Label	Admissible interpretation
Accepted BidKV paper/results	historical real-online, single device	Establishes the inherited local baseline only.
MultiNPU M0 document	design/contract	Freezes ownership questions, replay baselines, invariants, and negative stops.
Dedicated feature branches	unreviewed carrier	Identify future integration locations; no mechanism or result is admitted.
Deterministic replay	TBD simulation	Must create local/global divergence and pass accounting before runtime work.
Multi-NPU campaign	TBD	Required for any latency, throughput, scaling, or overhead statement.

Table 2: No historical single-device result is promoted to MultiNPU evidence.

7 Matched Evaluation Contract

7.1 Replay gate

The deterministic model must include at least two devices, asymmetric pressure, shared requests, independent local BidKV, the strongest non-BidKV local policy, global largest-first, and an offline oracle. It must contain a case where the local optimum is globally worse. If the treatment cannot beat local BidKV on any such case, runtime implementation stops.

7.2 Runtime identity and correctness

Any later vLLM-HUST execution uses the project-approved .23 official container. Host, devices, topology, model, container digest, parent and submodule heads, graph mode, command, requests, and result root come from a scheduler/allocation manifest; no fixed device or personal path is allowed.

The oracle requires: one live ownership epoch per decision; no request both resident and released; exact per-owner freed-block conservation; one attribution of recompute and transfer cost; no duplicate victim; no stale decision applied; and token-identical deterministic output versus the inherited local baseline. Missing receipts fail closed to local BidKV. Any correctness failure blocks performance claims.

7.3 Metrics and gates

Across at least three independent service starts, report task/request TTFT median/p95/p99, SLO attainment, throughput, preemptions, recomputed tokens, cross-device bytes, per-device freed blocks, coordination messages and latency, decision staleness, local/global divergence, oracle regret, fairness, failures, and all fallbacks. Freeze numerical efficacy and overhead thresholds before opening results. The minimum qualitative gate is a repeated deployable winner switch plus oracle headroom over independent local BidKV; an isolated latency delta is insufficient.

Stop if ownership cannot be observed precisely, coordination cost is comparable to avoided recomputation, choices rarely diverge, or gains arise from changing memory, requests, output length, execution mode, or correctness. Negative results close this mechanism, not the broader ownership problem.

Week one delivers the ownership map, typed replay, three baselines, oracle, and invariant tests. Week two names native hooks and implements a reviewable receipt path only if replay clears the action-space gate; hardware execution is a later, separately allocated student deliverable.

8 Related-Work and Claim Boundary

The accepted BidKV work supplies the strongest inherited local baseline [1]; its utility signal and results are not new MultiNPU evidence. PagedAttention documents the serving memory substrate and native reclamation context [3]. Dominant Resource Fairness formalizes multi-resource allocation fairness [2], but it does not bind logical requests to distributed KV ownership or price duplicated recomputation. The proposed gap is therefore not generic global scheduling. It is the conjunction of typed KV ownership, pressure-local decisions, and fail-closed coordination economics. A broader bibliography and direct multi-device victim-selection comparison are required before submission.

9 Conclusion

BidKV-MultiNPU is a separate research question, not a scaling label for an accepted single-device system. At the current head there is no replay counterexample, native mechanism, or hardware evidence. The project advances only if ownership-aware coordination produces observable action differences, passes exact accounting, and beats local BidKV after coordination cost. A clean negative boundary is preferable to relabeling inherited results.

References

- [1] Yanbo Chen, Mingqi Wang, Shuhao Zhang, Xiaofei Liao, and Hai Jin. Bidkv: Utility-guided preemption scheduling for kv-pressure llm serving. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 2026. Accepted single-device paper recorded by the repository README.
- [2] Ali Ghodsi, Matei Zaharia, Benjamin Hindman, Andy Konwinski, Scott Shenker, and Ion Stoica. Dominant resource fairness: Fair allocation of multiple resource types. In *Proceedings of the 8th USENIX Symposium on Networked Systems Design and Implementation*. USENIX Association, 2011.
- [3] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pages 611–626. ACM, 2023.